Marks Scored:

KATHMANDU UNIVERSITY
End Semester Examination
August/September, 2016

Level: B. E/ B.Sc./B. Tech. Course : MATH 104
Year : I Semester: II

Exam Roll No. : Time: 30 mins. F. M. : 20

Registration No.: Date :

SECTION "A"
[10 Q. × 1 = 10 marks]

1. The Cartesian form of r = Cos θ is ____________________________

2. If $\vec{F}$ is a field defined on D and $\vec{F} = \overrightarrow{\nabla f}$ for some scalar function f on D then f is called ________________________________________________________

3. The curl of a vector field $\vec{F}$ = M $\vec{\imath}$ +**N** $\vec{\jmath}$ at a point $(x_0, y_0)$ is ______________

4. The Fourier coefficient $a_n$ for the Fourier series $f(x) = a_0 + \sum_{n=1}^{n}(a_n\, Cosnx + b_n Sinnx)$ is given by $a_n$= ______________________________________________

5. The rate at which a fluid is entering or leaving a region enclosed by a smooth curve in the *xy*-plane is called______________________________________________

6. $\int_a^b \sqrt{(\frac{dx}{dt})^2 + (\frac{dy}{dt})^2 + (\frac{dz}{dt})^2}\ dt$ computes___________________________________

7. Change of $\int\int f(x,y)dxdy$ into its polar is___________________________________

8. The differential $P\ dx + Q\ dy + R\ dz$ is exact if and only if $\frac{\partial R}{\partial y} = \frac{\partial Q}{\partial z}$ , $\frac{\partial P}{\partial z} = \frac{\partial R}{\partial x}$ and ____________________________________________

9. If $f(x,y)$ is the function defined on $\Re$ containing the point $(a,b)$ then $f(x,y)$ has a saddle point at $(a,b)$ is if_______________________________________

10. The value of Γ (5/2) =______________________, the symbol has usual meaning.

SECTION "B"
[10 Q × 1 = 10 marks]

11. The curve $r^2 = 2a\ cos3\theta$ is symmetric about ________________________________________

[Pole, initial line, pole and initial line, line perpendicular to initial line]

12. $\int_0^1 \int_0^2 \int_0^3 dz\ dy\ dx$ = ________________________________

13. The vector field $\boldsymbol{F} = M\ \vec{i} + N\,\vec{j}\ + P\vec{k}$ is conservative if $\frac{\partial M}{\partial y} = \frac{\partial P}{\partial x}, \frac{\partial M}{\partial z} = \frac{\partial N}{\partial x}$, and

________________________________

14. The polar conic section defined as $r = \frac{ek}{1-e\cos\theta}$ has equation of directrix for ______

[$x = k$, $x = -k$, $y = k$, $y = -k$ ]

15. The spherical coordinate system is ________________________________

[$(\rho, \varphi, \theta)$, $(\theta, \rho, \varphi,)$, $(r, \rho, \varphi,)$, $(\theta, \rho, r,)$]

16. The equation of tangent plane to surface $f(x, y, z) = C$ at $P_0(x_0, y_0, z_0)$ is

________________________________

(i) $f_x(P_0)(x - x_0) + f_y(P_0)(y - y_0) + f_z(P_0)(z - z_0) = 0$

(ii) $f_x(P_0)(x - x_0) + f_y(P_0)(y - y_0) + (z - z_0) = 0$

(iii) $(x - x_0) + f_y(P_0)(y - y_0) + f_z(P_0)(z - z_0) = 0$

(iv) $f_x(P_0)(x - x_0) + (y - y_0) + f_z(P_0)(z - z_0) = 0$

17. If $w = f(x, y), x = g_1(t)$ , $y = g_2(t)$ , $\frac{dw}{dt} =$ ________________________________

(i) $\frac{\partial x}{\partial t}\ \frac{dw}{dx} + \frac{\partial y}{\partial t}\ \frac{dw}{dy}$

(ii) $\frac{\partial x}{\partial t}\ \frac{\partial w}{\partial x} + \frac{\partial y}{\partial t}\ \frac{\partial w}{\partial y}$

(iii) $\frac{\partial x}{\partial t}\ \frac{dt}{dx} + \frac{\partial w}{\partial t}\ \frac{dt}{dy}$

(iv) $\frac{\partial w}{\partial x}\ \frac{dw}{dt} + \frac{\partial w}{\partial y}\ \frac{dy}{dt}$

18. $r = 2a\ Sin\theta$ ________________________________

(i) Cylinder (ii) Cardiod (iii) Rose petal (iv) Circle

19. $\lim_{(x,y)\to(2,2)} \frac{x+y-4}{\sqrt{x+y}-2} =$ ________________________________

[ 2, 4 , 6 , 12]

20. $\int_0^{\frac{\pi}{2}} cos^6\theta\ d\theta =$ ________________________________

[ $\frac{1}{2}$, $\frac{3}{2}$ , $\frac{5}{2}$, 6 ]

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| Time : 2 hrs. 30 mins. | F. M. : 55 |

SECTION "C"
[4 Q. × 7 = 28 marks]

Attempt *ALL* questions.

1. What is a Fourier series? Find the Fourier coefficients $a_0$ , $a_n$, $b_n$ for the following piecewise continuous function: [1+2+2+2]

$$f(x)=\begin{cases} 0 & if -\pi<x<0 \\ x & if\ 0<x<\pi \end{cases}$$

2. Define curvature and torsion for a space curve. Also find curvature and torsion for the curve $\vec{r}(t) = a\,cost\,\vec{\imath} + a\,sint\,\vec{\jmath} + bt\,\vec{k}$ where $a, b > 0$ [2+2.5+2.5]

OR

Define $\vec{T}$ , $\vec{N}$ , $\vec{B}$ and find them for the curve $\vec{r}(t) = e^t\,cost\,\vec{\imath} + e^t\,sint\,\vec{\jmath} + 2\,\vec{k}$ , the symbols have their usual meanings. [1+2+2+2]

3. State the Second Derivative Test for evaluating local maxima, local minima, critical points, and saddle points for a function $f(x, y)$. Also, use this method to find local extrema of $f(x, y) = x^3 + y^3 + 3x^2 - 3y^2 - 8$ . [2+5]

4. Write necessary conditions for Green's theorems to hold. Also, verify the normal form of Green's theorem for the vector field $\vec{\boldsymbol{F}} = -\boldsymbol{x^2}\,\vec{\boldsymbol{\imath}} + \mathbf{x}\boldsymbol{y^2}\,\vec{\boldsymbol{\jmath}}$ on the bounding circle $\vec{r}(t) = a\,cost\,\vec{\imath} + a\,sint\,\vec{\jmath}$ , $0 \le t \le 2\pi$ [2+5]

SECTION "D"
[9Q × 3 = 27 marks]

Attempt *ALL* questions.

5. Find the equation of the plane tangent to the surface $z = e^{(x^2+y^2)}$ at $(0, 0, 1)$.

OR

Find the directional derivative of $f(x, y, z) = x^2 - y^2x - z$ at $P(1, 1, 0)$ in the direction of $\vec{A} = 2\vec{\imath} - 3\vec{\jmath} + 6\vec{k}$ .

6. Evaluate $\int_0^2 \int_{-\sqrt{4-y^2}}^{\sqrt{4-y^2}} \int_0^{2-x-y} dzdxdy$ .

7. State the Gamma function and use it to evaluate $\int_0^{\pi/2} sin^6\theta \, . cos^7\theta\, d\theta$. [1+2]

8. Find the flow along the helix $\vec{r}(t) = Cost\,\vec{\imath} + Sint\,\vec{\jmath} + t\,\vec{k}$ ; $0 \le t \le \frac{\pi}{2}$ for a fluid's velocity field $\vec{F} = x\vec{\imath} + z\vec{\jmath} + y\vec{k}$ .

9. Find the length of Cardoide $r = 2 + 2cos\theta$.

10. Solve the initial value problem: $\frac{d\vec{r}}{dt} = (t^3 + 4t)\vec{i} + t\,\vec{j} + 2t^2\,\vec{k}$ with $\vec{r}(0) = \vec{i} + \vec{j}$.

11. Write the geometrical meaning of first order partial derivative of $f(x, y, z)$. Also, if w = $x^2 + y - z$ + Sint, $x + y = $ t then find $(\frac{\partial w}{\partial t})_{x\,,z}$ . [1+2]

12. Find the potential function for the conservative field of the vector field $\overrightarrow{F} = (y + z)\vec{i} + (x + z)\vec{j} + (x + y)\vec{k}$.

13. Convert polar coordinate system $(2,\ \frac{2\pi}{3},\ 1)$ into spherical and cylindrical coordinate system. [1.5+1.5]